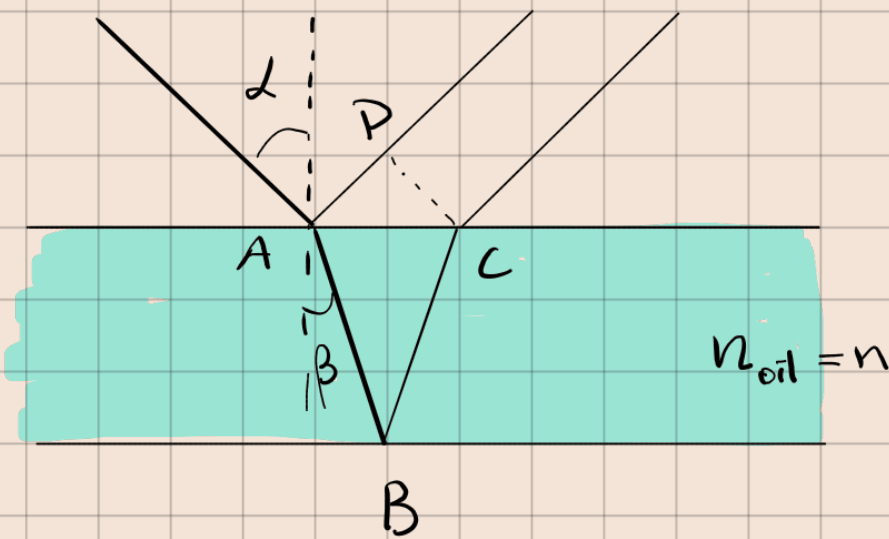


Ex 29



Calculate the optical path difference.

$$\Delta s = n \cdot (\overline{AB} + \overline{BC}) - \underbrace{n_{air}}_{=1} \cdot \overline{AD}$$

$$\overline{AB} = \overline{BC} = \frac{d}{\cos \beta}$$

$$\begin{aligned} \overline{AD} &= \overline{AC} \cdot \sin \alpha = 2 \cdot \sin \beta \cdot \sin \alpha \cdot \overline{AB} \\ &\quad \underbrace{2 \cdot \sin \beta \cdot \overline{AB}}_{=2 \cdot d \cdot \frac{\sin \beta}{\cos \beta}} = 2 \cdot d \cdot \frac{\sin \beta}{\cos \beta} \cdot \sin \alpha \end{aligned}$$

$$\Rightarrow \Delta s = \frac{2d}{\cos \beta} \cdot (n - \sin \beta \cdot \sin \alpha)$$

Refraction law: $n \cdot \sin \beta = \sin \alpha$

$$\Rightarrow \Delta s = \frac{2d}{\sqrt{\frac{n^2}{n^2} - \frac{\sin^2 \alpha}{n^2}}} \left(\frac{n^2}{n} - \frac{\sin^2 \alpha}{n} \right)$$

$$= 2d \sqrt{n^2 - \sin^2 \alpha} = 2dn \sqrt{1 - \sin^2 \beta} \\ \underbrace{n^2 \cdot \sin^2 \beta}_{= 2dn \cdot \cos \beta}$$

For constructive interference,

$$\Delta s = \left(m + \frac{1}{2}\right) \lambda$$

insert $m=0$, $\lambda=500 \text{ nm}$, $\alpha=45^\circ$

$$\Rightarrow d = \frac{\lambda}{4 \sqrt{n^2 - \sin^2 \alpha}} = 8.7 \times 10^{-8} \text{ m}$$

Exercise 30

Convex lens formula:

$$\frac{1}{f} = \frac{1}{b} + \frac{1}{g} \quad \left(\Rightarrow \frac{1}{b} = \frac{1}{f} - \frac{1}{g} \right)$$

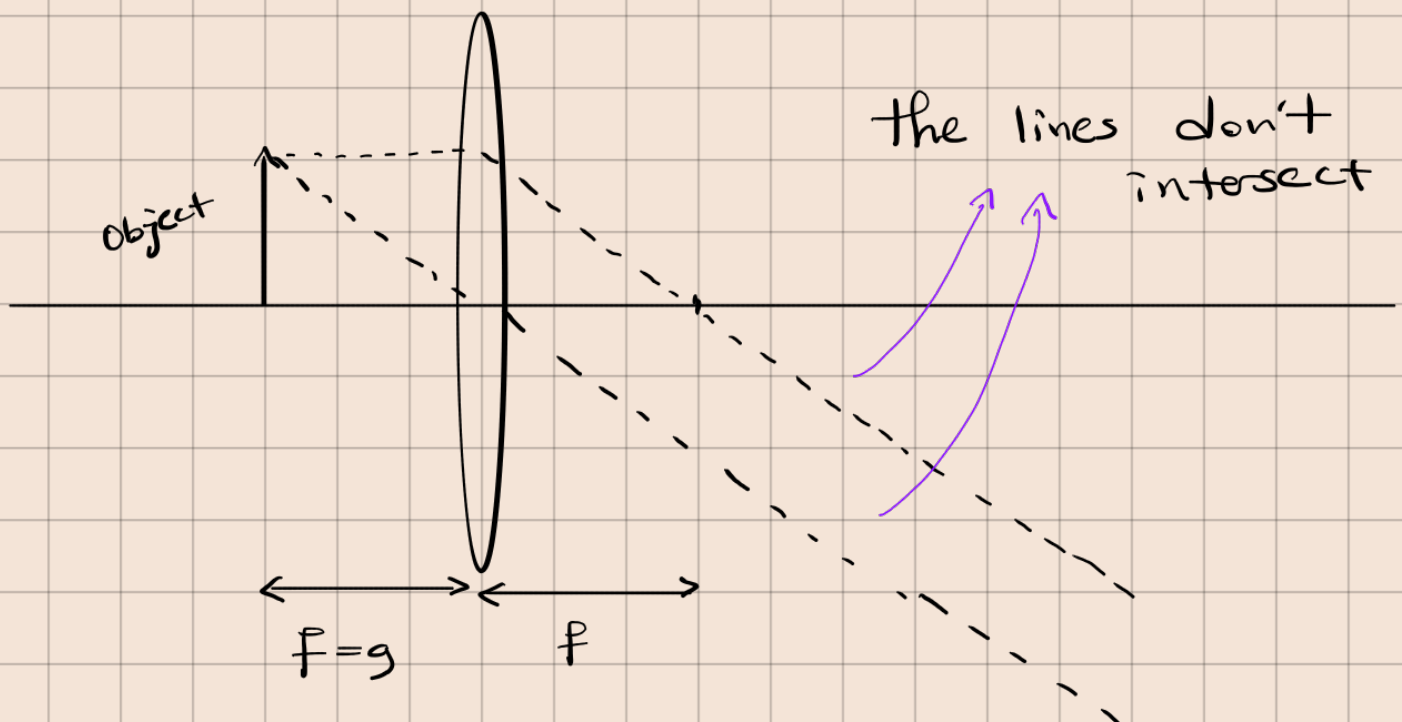
$\downarrow$ focal length $\hookrightarrow$ image $\hookrightarrow$ object

Here, $f = 30 \text{ mm}$.

a) $g = f = 30 \text{ mm}$

$$\Rightarrow \frac{1}{b} = \frac{1}{f} - \frac{1}{f} = 0 \Rightarrow b \rightarrow \infty$$

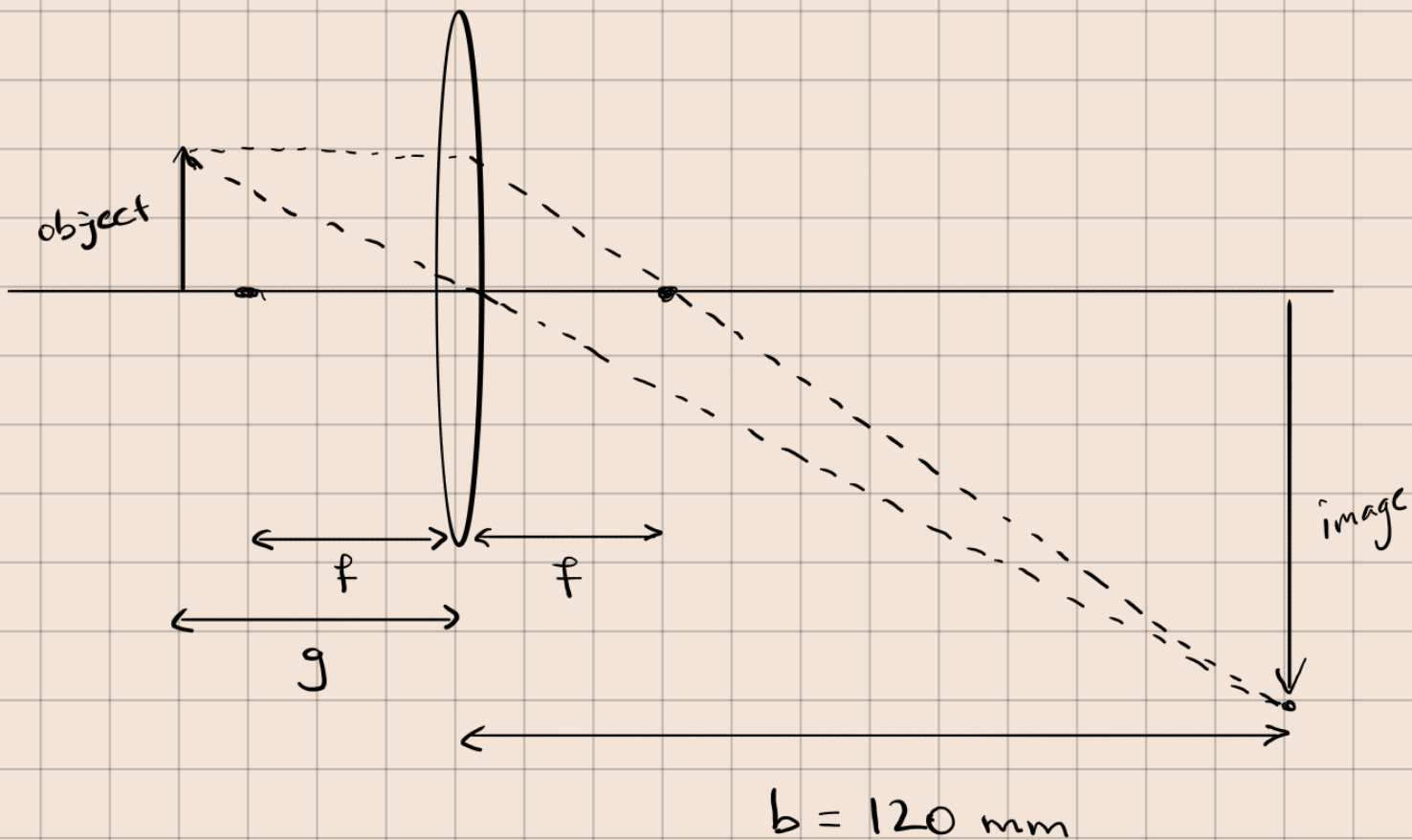
There is no image.



b) $g = 40 \text{ mm}$

$$\Rightarrow \frac{1}{b} = \frac{1}{30 \text{ mm}} - \frac{1}{40 \text{ mm}} \Rightarrow b = 120 \text{ mm}$$

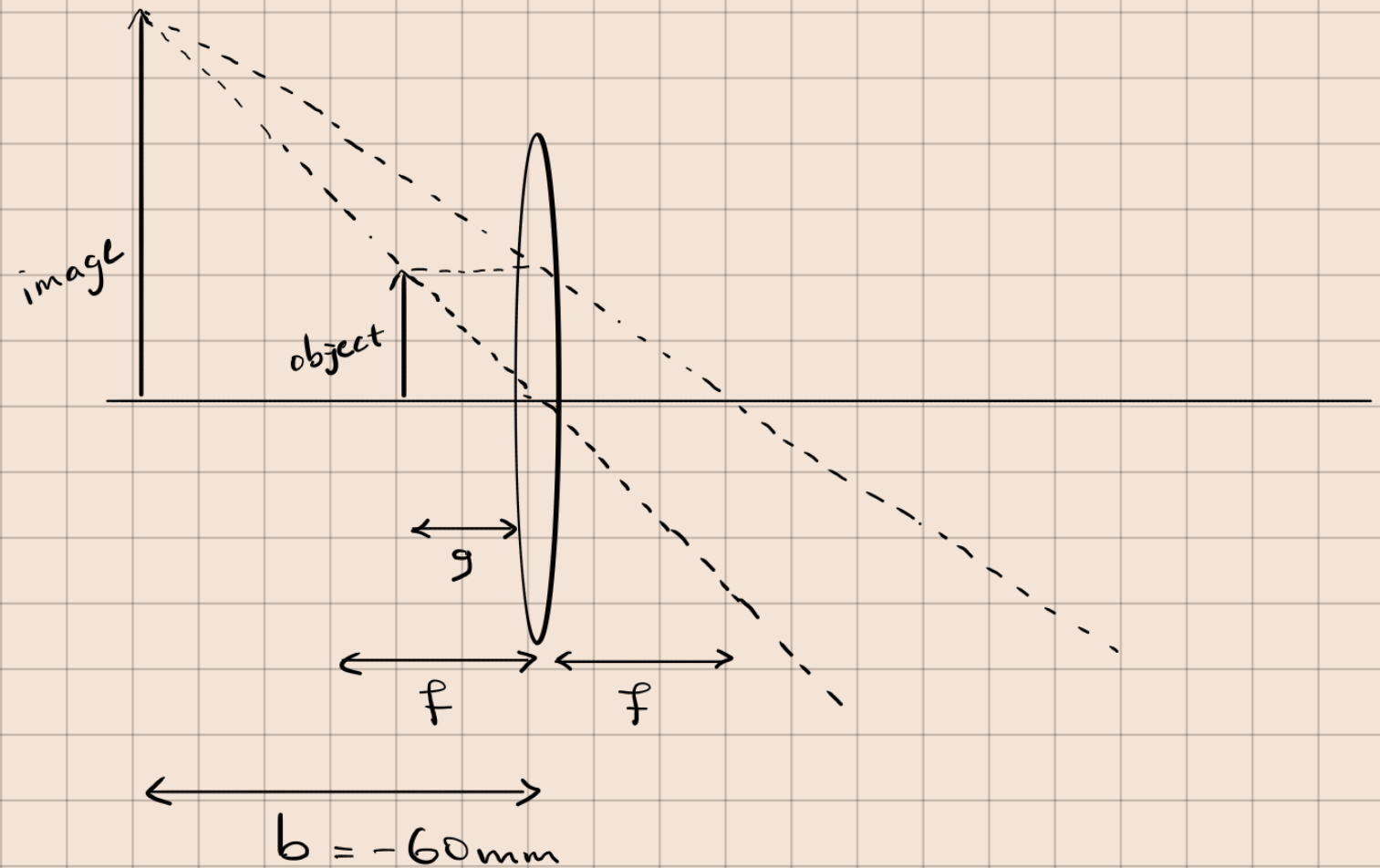
Real image.



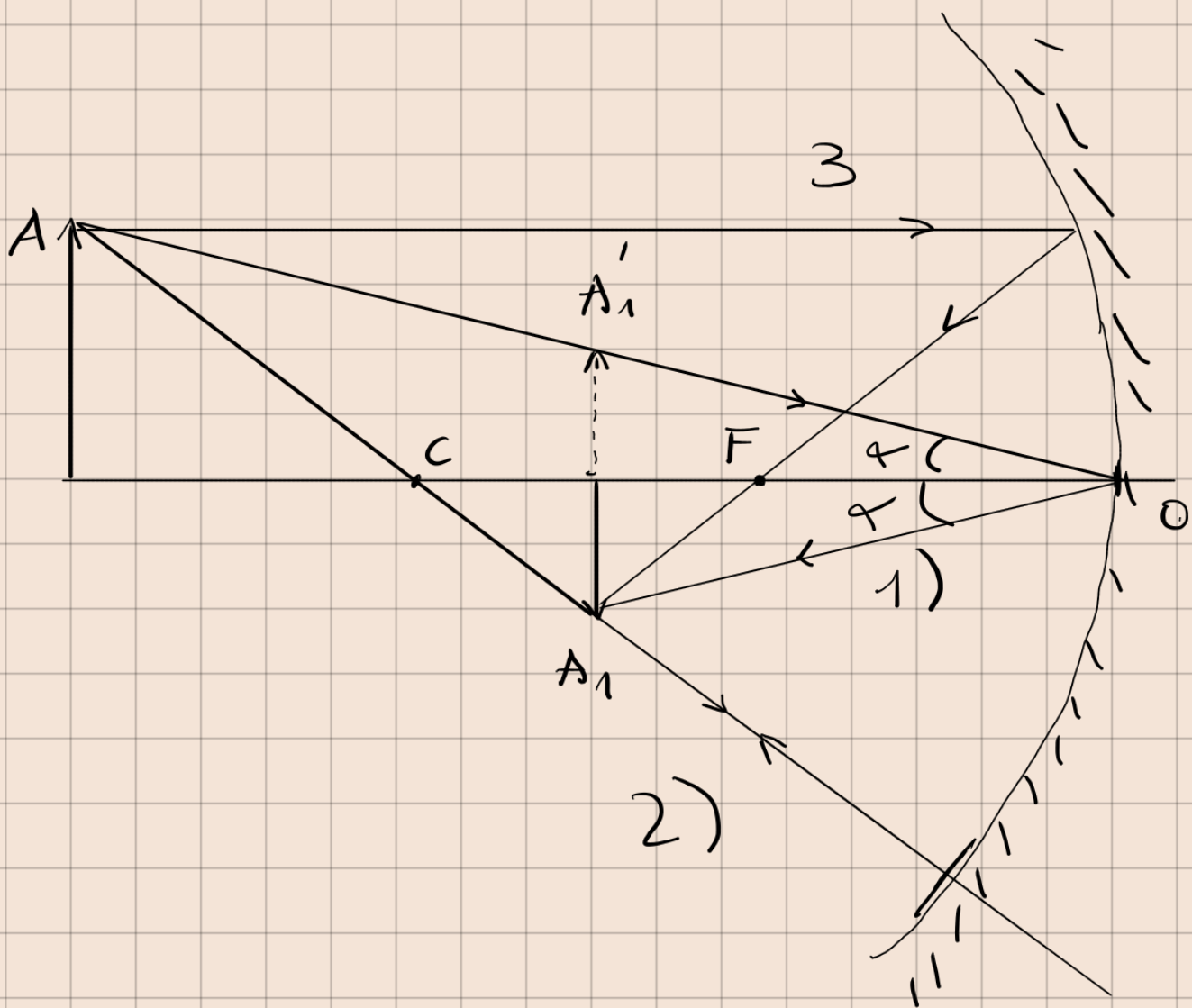
c) $g = 20 \text{ mm}$

$$\Rightarrow \frac{1}{b} = \frac{1}{30 \text{ mm}} - \frac{1}{20 \text{ mm}} \Rightarrow b = -60 \text{ mm}$$

Virtual image



Ex 31



- 1) Ray passing through A and A_1' must be reflected with the same angle from O (the vertex of the mirror).

2) Ray passing through A and A_1 must be reflected on itself.

→ This gives us the center of curvature C .

3) The focal point can be found from $f = \frac{r}{2}$.

We can also draw the focal ray going through F .

Ex 32

$$a) \quad \frac{1}{f} = \frac{1}{g} + \frac{1}{b} \quad \text{with} \quad f = 0.12 \text{ m} \\ b = 8 \text{ m}.$$

$$\Rightarrow \frac{1}{g} = \frac{1}{f} - \frac{1}{b}$$

Magnification:

$$\gamma = \frac{B}{G} = \frac{b}{g} = b \left(\frac{1}{f} - \frac{1}{b} \right) \\ = \frac{b}{f} - 1 = 65.67$$

Slide dimensions are

$$h = 24 \text{ mm}, \quad w = 36 \text{ mm}.$$

After magnification, image dimensions are

$$H = \gamma \cdot h = 1.57 \text{ m}$$

$$W = \gamma \cdot w = 2.36 \text{ m} > 2 \text{ m}$$

⇒ Screen is too small.

b) We increased b from 8m to 8.3 m.

$$\frac{1}{f} = \frac{1}{g} + \frac{1}{b} = \text{constant}$$

b gets larger $\Rightarrow 1/b$ gets smaller

$\Rightarrow 1/g$ must get larger

$\Rightarrow g$ must get smaller

$\Rightarrow$ The distance between slides and lens must be reduced.

c) Now: $b = 3 \text{ m}$

Use the formula for magnification:

$$\gamma = \frac{b}{f} - 1$$

$$W = \gamma \cdot w = \left(\frac{b}{f} - 1 \right) \cdot w$$

$$W(f = 45 \text{ mm}) = 2.34 \text{ m} \quad (\text{too big})$$

$$W(f = 60 \text{ mm}) = 1.76 \text{ m}$$

$$W(f = 90 \text{ mm}) = 1.16 \text{ m} \quad (\text{small})$$

60 mm gives the best result.